You must show your work to receive credit.

1. $\sqrt{3x+18} = x$

$$\begin{aligned}\left(\sqrt{3x+18}\right)^2 &= x^2\\ 3x+18 &= x^2\\ 0 &= x^2-3x-18\\ 0 &= (x-6)(x+3)\end{aligned}$$

So $x = 6$ and $x = -3$ are *possible* answers. Checking we find that

$$x = -3 \Rightarrow \sqrt{9} = 3 \neq -3 \text{ so } x = -3 \text{ is NOT an answer, and}$$

$$x = 6 \Rightarrow \sqrt{36} = 6\checkmark$$

So $x = 6$.

2. $6x^{5/2} - 12 = 0$

$$\begin{aligned}6x^{5/2} &= 12\\ x^{5/2} &= \frac{12}{6}\\ \left(x^{5/2}\right)^{2/5} &= (2)^{2/5}\\ x &= 2^{2/5} = \sqrt[5]{2^2} = \sqrt[5]{4}\end{aligned}$$

Note that: $\left(x^{5/2}\right)^{2/5} = x^{5/2 \cdot 2/5} = x^1$.

3. $|x+1| + 5 = 3$

$$|x+1| = -2$$

We need to solve for two cases

$$\begin{aligned}x+1 = 2; &\quad x+1 = -2\\ x = 1 &\quad\quad x = -2\end{aligned}$$

Checkin these answers we find that $|1+1| = 2 \neq -2$ and $|-2+1| = 1 \neq -2$. So there is no solution.